

Worksheet 1A: Fractions Fundamentals — Solutions

AIML Foundations Mathematics

Dublin and Dún Laoghaire ETB

Instructor: Josh Aaron

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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. $\frac{19}{12}$
2. $\frac{5}{24}$
3. $\frac{1}{6}$
4. 6
5. $\frac{3}{4}$
6. $\frac{2}{3}$
7. $\frac{4}{3}$
8. $\frac{22}{15}$
9. $\frac{7}{2}$ or $3\frac{1}{2}$
10. 4
11. 5
12. 21

Part B

13. $\frac{7}{4} = 1.75$
14. $\frac{15}{8} = 1.875$
15. $\frac{31}{16} = 1.9375$
16. Approaches 2
17. $\frac{3}{4}$ (telescopes: $1 - \frac{1}{4}$)
18. $\frac{4}{5}$ (telescopes: $1 - \frac{1}{5}$)
19. Approaches 1
20. $\frac{65}{24} \approx 2.708$

Part C

21. $\frac{2p}{3}$
22. $\frac{3m^2}{5n^2}$
23. $t - 2$
24. $\frac{r-3}{r+3}$
25. 2
26. $\frac{h-3}{h+2}$
27. n
28. $n + 1$
29. $\frac{(n+2)(n+1)}{2}$
30. $\frac{3s-2}{s+2}$

Part D

31. $\frac{3p+2q}{pq}$
32. $\frac{3t-7}{(t+1)(t-1)}$
33. $\frac{3n+2m-1}{mn}$
34. $\frac{r^2+4}{(r+2)(r-2)}$
35. $\frac{2}{k^2-1}$
36. $\frac{s-3}{2s}$
37. $w(w + 2)$ or $w^2 + 2w$
38. $\frac{3u^2+6u+2}{u(u+1)(u+2)}$
39. $\frac{p+1}{p+2}$
40. $k = \frac{mn}{m+n}$

Part E

41. $\frac{q+p}{q-p}$
42. $\frac{t-1}{3(t+1)}$
43. $\frac{n-1}{n}$
44. $\frac{m^2+n^2}{m^2-n^2}$
45. $\frac{k+1}{2k+1}$
46. $\frac{1}{h(h+1)}$

Part F

- 47. (a) 0.75 (b) 0.5625 (c) 0.4219 (d) 0.3164 (e) Approaches 0
- 48. (a) 1.25 (b) 1.5625 (c) 1.9531 (d) 2.4414 (e) Grows without bound
- 49. less, 0; greater, infinity
- 50. $\frac{15}{16} = 0.9375$
- 51. ≈ 0.349 — closer to 0.5 but below it
- 52. ≈ 0.366 — surprisingly close to $\frac{1}{e}$; even tiny reductions compound significantly

Part G

- 53. (a) Grows without bound (b) Approaches 0
 - 54. (a) Approaches 0 (b) Grows without bound
 - 55. (a) $1 + \frac{1}{k}$ (b) Approaches 1
 - 56. (a) $\frac{1}{2}$, $\frac{100}{101} \approx 0.99$, $\frac{10000}{10001} \approx 0.9999$ (b) Approaches 1 (c) Never reaches it
 - 57. (a) Approaches 1 (b) Approaches 0
 - 58. (a) Approaches 1 (b) Always less than 1
 - 59. (a) Stays the same (b) Depends: increases if $\frac{a}{b} < 1$, decreases if $\frac{a}{b} > 1$
 - 60. (a) 2, 1.1, 1.01 (b) Approaches 1 (c) $e \approx 2.718$
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